STROBE-MR Checklist

Study: Genetically higher bone mineral density raises knee osteoarthritis but protects against fracture: Mendelian randomisation of 13 skeletal phenotypes
Reporting guideline: STROBE-MR (Skrivankova et al., BMJ 2021;372:n223).

#	STROBE-MR item	Status	Location in manuscript
1	Title — identifies the report as a Mendelian randomisation study	Reported	Title: "…Mendelian randomisation of 13 skeletal phenotypes"
2	Abstract — follows STROBE-MR abstract structure (background, objectives, methods, results, conclusions)	Reported	Abstract: background/objectives/methods/results/conclusions present (unstructured per OAC)
3	Introduction – Background: rationale for the MR investigation	Reported	§1 Introduction (Foss & Byers 1972 → present debate; why MR)
4	Introduction – Objectives: specific objectives/hypotheses	Reported	§1 final paragraph (three open questions; 13-phenotype design)
5	Methods – Data sources: GWAS provenance, search, eligibility, date accessed	Reported	§2 Data sources (13 GWAS; Refs 1–3,10; accessions)
6	Methods – GWAS description: phenotype definition, participants, QC, effect allele, units, population	Reported	§2 Data sources; Table 1 (h^2, N, intercepts); §3 heritability
7	Methods – Exposure instrument: selection, number, pruning/clumping, population, proxies, palindromes	Reported	§2 MR ($p<5\times10^{-8}$; $r^2<0.001$/10 Mb; 20–35 per BMD; 116 knee OA; HapMap3 EUR)
8	Methods – Outcome instrument / outcome GWAS: same detail	Reported	§2 Data sources (knee-OA GO 2025 N≈1.21M; self-rep IEU ukb-b-14486; fracture GEFOS)
9	Methods – Assumption 1 (relevance): instrument–exposure association strength (F, R^2)	Reported	§2 MR (instrument counts; Table 3; S9 conditional F ≥10); §2 Effect-scale reconstruction
10	Methods – Assumption 2 (independence): pleiotropy assessment plan	Reported	§2 MR (MR-Egger intercept, MR-PRESSO, leave-one-out); §3 robustness
11	Methods – Assumption 3 (exclusion restriction): no confounders of instrument–outcome	Reported	§2 Sample overlap (Burgess bound); §2 MRlap; §4 limitations (residual overlap)
12	Methods – Statistical analysis: MR estimators, heterogeneity, sensitivity, software	Reported	§2 MR/LDSC/Colocalisation/MVMR (IVW, MR-Egger, weighted median, MR-PRESSO, LOO; Python 3.13; MRlap R 4.6.1)
13	Methods – Sample overlap: assessment and handling	Reported	§2 Sample overlap (Burgess approximation; bound 2.6% BMD→OA, 12% BMD→fracture; MRlap)
14	Results – GWAS descriptives: N, cases, h^2, intercept, population	Reported	Table 1; §3 Genetic-correlation architecture (intercepts 1.01–1.07)
15	Results – Exposure instrument: SNPs, F-statistic, R^2	Reported	§2 MR (counts); Table 3 (n_snp); S9 (Fc≥10)
16	Results – Outcome instrument: same	Reported	Table 3 (knee OA 69 SNPs); §2 Data sources
17	Results – Assumption testing: heterogeneity (Cochran Q/I^2), pleiotropy (MR-Egger, MR-PRESSO), sensitivity	Reported	§3 (I^2 68–86%; MR-Egger intercept p>0.24; MR-PRESSO 9/10 significant, 0–3 outliers); Table 3
18	Results – Main MR estimates: with CIs, p-values, direction	Reported	Table 3 (IVW β, SE, p, random-effects p, Egger, weighted median, I^2, LOO); §3
19	Results – Supplementary/stratified analyses: MVMR, coloc, reverse, MRlap	Reported	§3 (positive control, reverse MR, coloc, MVMR S9, MRlap Fig 10); S8, S9, S10
20	Results – Sample overlap results: empirical overlap diagnostics	Reported	§3 (Fig 9 ρ_GE; §2 MRlap median +20.5% away from null)
21	Discussion – Key results: interpreted in context	Reported	§4 Principal findings; Resolving the OA–BMD paradox; Site and direction
22	Discussion – Limitations, interpretation, generalisability, conclusions, funding, ethics	Reported	§4 Strengths and limitations; Conclusion; Declarations (Funding: None; Competing interests: none); Data availability

Overall: all 22 STROBE-MR items are addressed in the main manuscript and accompanying supplementary files. No item is Not Applicable.